

Automated Student Lab Allocation Using Maximum Bipartite Matching

OUTPUTS:

LabMatch
Smart Lab Allocation

Dashboard

Import Data

Run Allocation

Seating Plan

Find Student

Branch Analysis

Reports

"Efficient allocation
for a smarter tomorrow"

Backend online

NG N. Greeshma
Admin

Dashboard

Smart student-to-lab allocation using maximum bipartite matching.

Students
0

Labs
0

Total Seats
0

Allocated
0

LabMatch Allocation System

Import your student and laboratory records, then generate an optimized seating allocation.

Import Data

View Seating Plan

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Import Data

Upload student records and laboratory information.

Student Records

Import your student data. The system analyses the uploaded records and extracts the available student information automatically.

Choose Student File

✓ 100 student records imported

Laboratory Data

Import laboratory names and available seating capacity. The allocation system will use the available capacity when generating the matching.

Choose Lab File

✓ 9 laboratories imported

Imported Data Summary

Student Records
100

Laboratories
9

Available Seats
362

9 laboratories imported successfully.

Final Allocation

Generated student-to-seat assignments using maximum bipartite matching.

Students

100

Total Capacity

362

Allocated

100

Unallocated

0

Student-to-Seat Assignments

Each student is assigned to at most one available seat.

Reallocate

ROLL NO	STUDENT	BRANCH	ALLOCATED LAB	SEAT	STATUS
STU001	Aarav Sharma	CSE	14A	14A-01	Allocated
STU002	Ananya Reddy	CSE	10B	10B-01	Allocated
STU003	Arjun Kumar	CSE	14B	14B-46	Allocated
STU004	Aditi Rao	ECE	14B	14B-35	Allocated
STU005	Aditya Verma	ECE	HC-01	HC-01-30	Allocated
STU006	Akshara Singh	EEE	15A	15A-13	Allocated
STU007	Aman Gupta	EEE	301	301-08	Allocated
STU008	Amrutha Reddy	IT	15B	15B-03	Allocated
STU009	Aniket Sharma	IT	303	303-34	Allocated
STU010	Anjali Nair	ME	008	008-45	Allocation complete: 100/100 students assigned.
STU011	Amanu Datta	CSE	HC-01	HC-01-A1	Allocated

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Seating Plan

Visualize assigned students inside each laboratory.

Select Lab

14B

Lab Name

14B

3D View

Assigned

11

Total Seats

49

14B

AVAILABLE

STU041

AVAILABLE

AVAILABLE

STU001

AVAILABLE

AVAILABLE

AVAILABLE

AVAILABLE

AVAILABLE

AVAILABLE

AVAILABLE

AVAILABLE

AVAILABLE

AVAILABLE

STU071

AVAILABLE

AVAILABLE

STU037

AVAILABLE

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NGN. GreeshmaAdmin

Find Student

Search for a student and locate their laboratory and seat.

Aksh

Search

Student Location

Roll Number

STU006

Student

Akshara Singh

Branch

EEE

Lab

15A

Seat

15A-13

Row

3

Column

3

Branch Analysis

Understand how students from different branches are distributed.

Students by Branch

CSE	30
ECE	20
EEE	20
IT	20
ME	10

Allocated Students by Branch

CSE	30
ECE	20
EEE	20
IT	20
ME	10

Reports

Allocation statistics and laboratory utilization.

Total Students

100

Total Seats

362

Allocation Rate

100.0%

Laboratory Utilization

LAB	CAPACITY	ASSIGNED	UTILIZATION	STATUS
14A	30	12	40%	Used
14B	49	11	22%	Used
15A	42	11	26%	Used
15B	30	11	37%	Used
008	48	11	23%	Used
108	30	11	37%	Used
HC 01	42	11	26%	Used
301	35	11	31%	Used
303	56	11	20%	Used